

CS & IT ENGINEERING



Computer Network

Routing

Lecture No. - 01



By - Abhishek Sir



Recap of Previous Lecture



Topic

ICMP

Topic

DHCP



Topics to be Covered



Topic

Routing



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : Syllabus



Concept of layering : OSI and TCP/IP Protocol Stacks;

Basics of packet, circuit and virtual circuit switching;

Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT);

Routing protocols : shortest path, flooding, distance vector and link state routing;



Topic : Forwarding



Data Plane :

- > Determine how IP packets are forwarded
[Using Router's Forwarding/Routing table]
- > Move packet from a router's input link to appropriate router's output link





Topic : Routing



Control Plane :

- > Determine how IP packets routed among routers
[Determine route taken by IP packets from source to destination]
- > Construction of Router's Forwarding/Routing table
[Using Routing algorithms/protocols]



Topic : Routing



→ Identify optimal (best) path between Source and Destination.

→ Types of Routing :

1. Static Routing

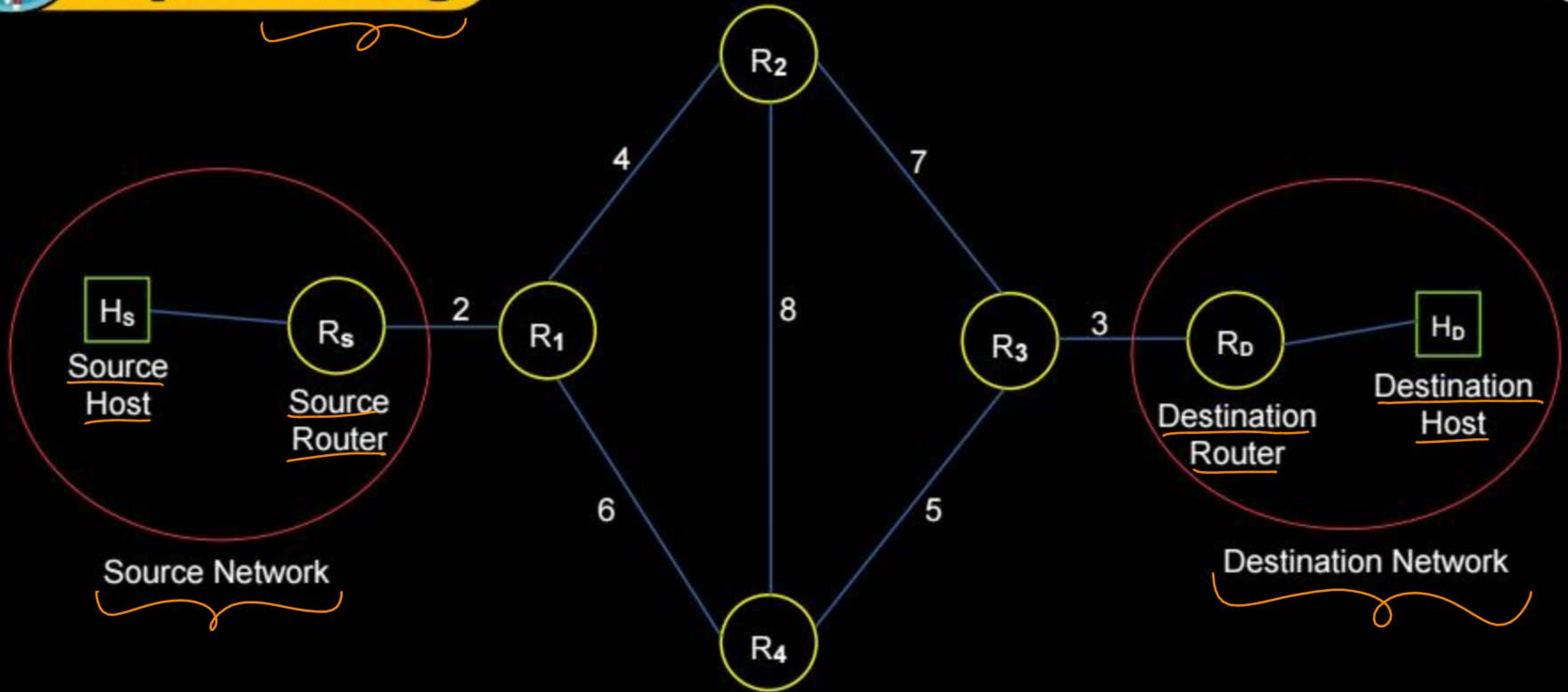
→ Non-adaptive Routing

2. Dynamic Routing

→ Adaptive Routing



Topic : Routing





Topic : Link Cost



→ Metric to compare link :

1. Distance (Hop Count)
2. Delay
3. Bandwidth

Cost $\propto$ Distance

Cost $\propto$ Delay

Cost $\propto \frac{1}{\text{Bandwidth}}$



Topic : Static Routing



→ Non-adaptive Routing

→ Types of Static Routing :

1. Shortest Path Algorithm
2. Flooding



Topic : Shortest Path Algorithm

=> Single-source shortest path algorithm

→ Dijkstra's algorithm ✓

→ Bellman-Ford algorithm

=> All-pairs shortest path algorithm

→ Floyd-Warshall algorithm



Topic : Dijkstra's Algorithm



→ Dijkstra's Algorithm is divided into two steps :

1. Initialization
2. Iterative distance calculation



Topic : Dijkstra's Algorithm



Graph
[Adjacency Matrix or List]

Source

Dijkstra
Algorithm

Shortest path to
all other nodes

Use as a
forwarding



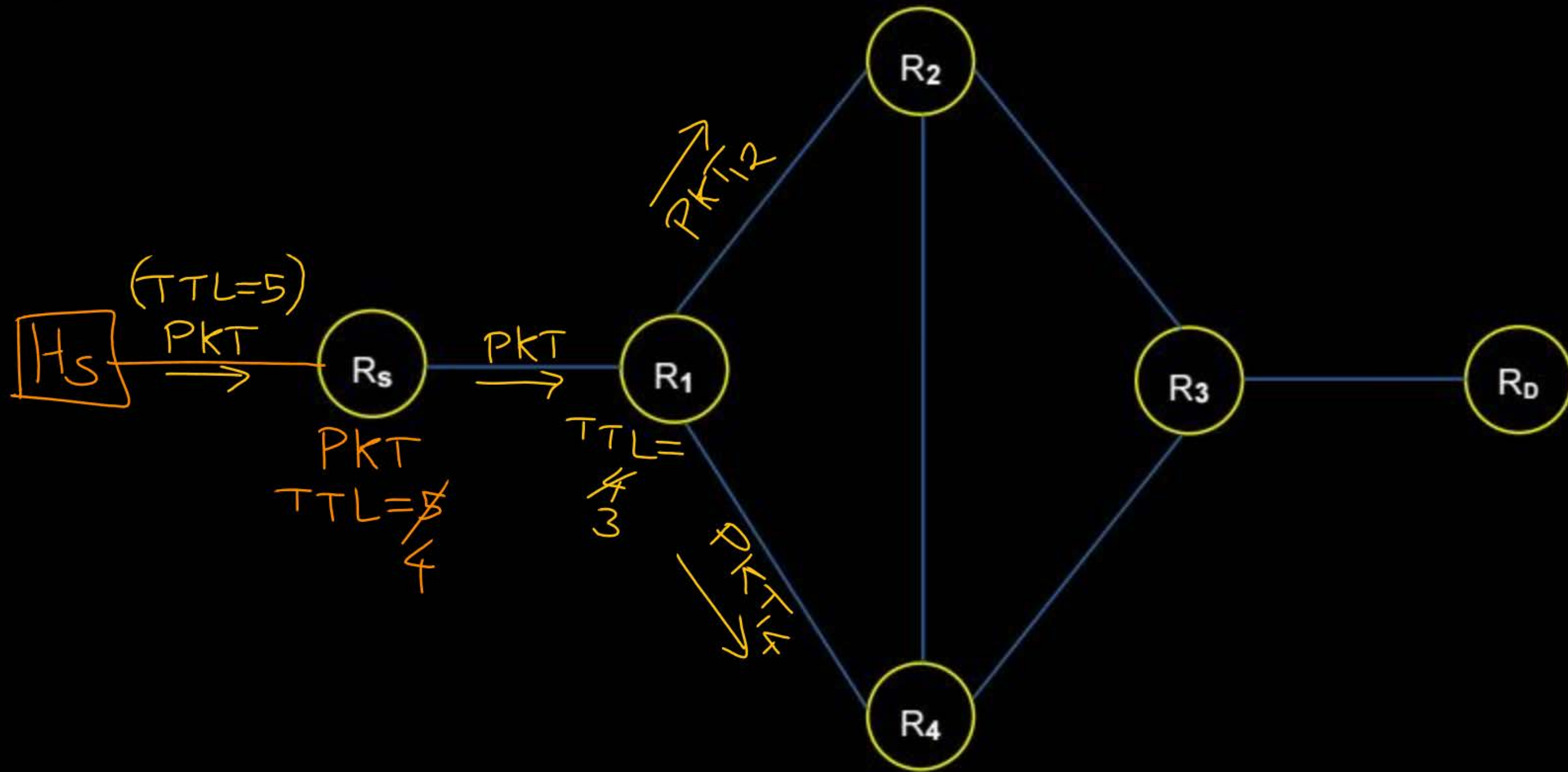
Topic : Flooding



- Router always broadcast arrived packet
[sent to all the outgoing links except incoming link]
- Vast number of packets in the network
- The packet which follow shortest path deliver first

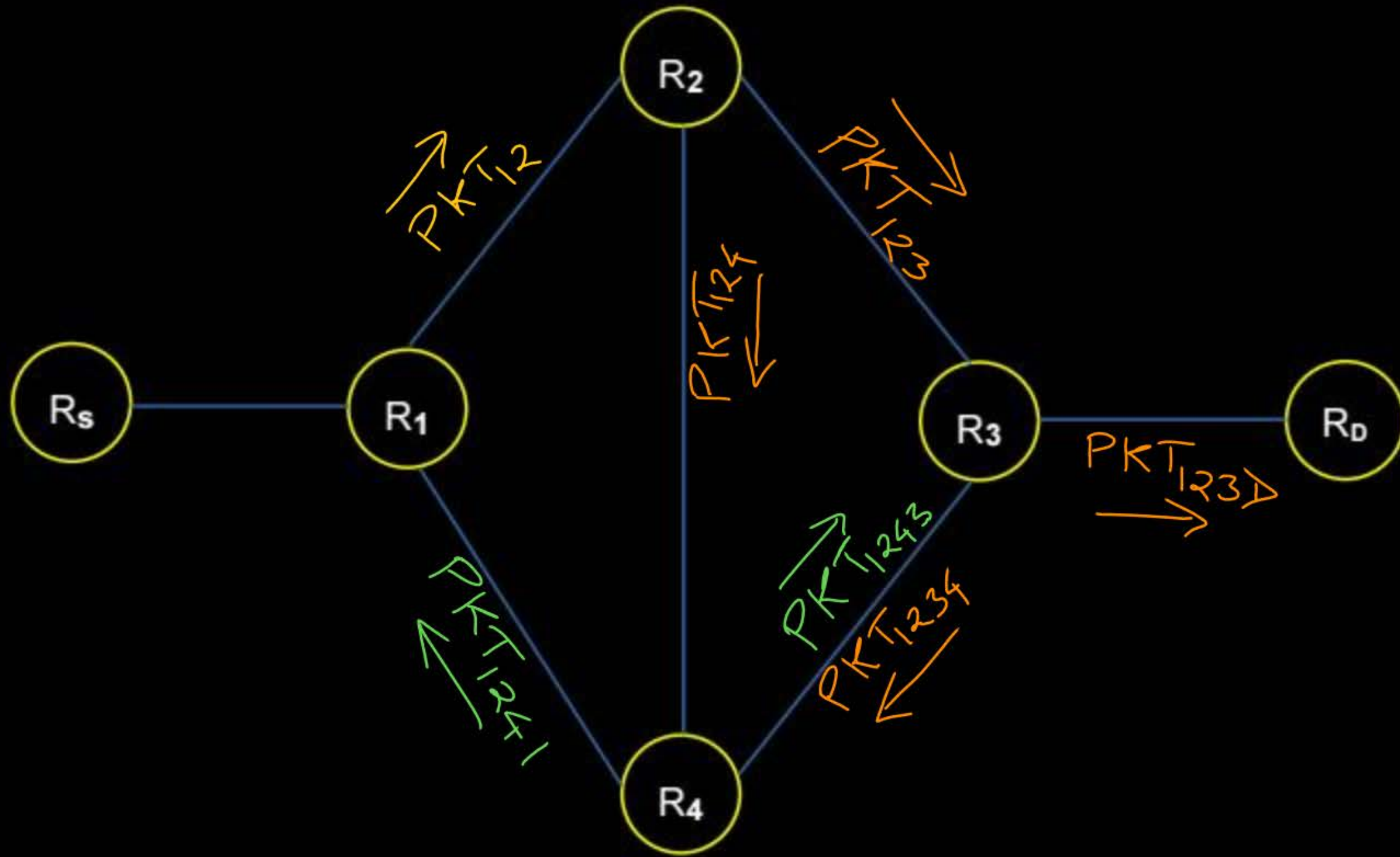


Topic : Flooding



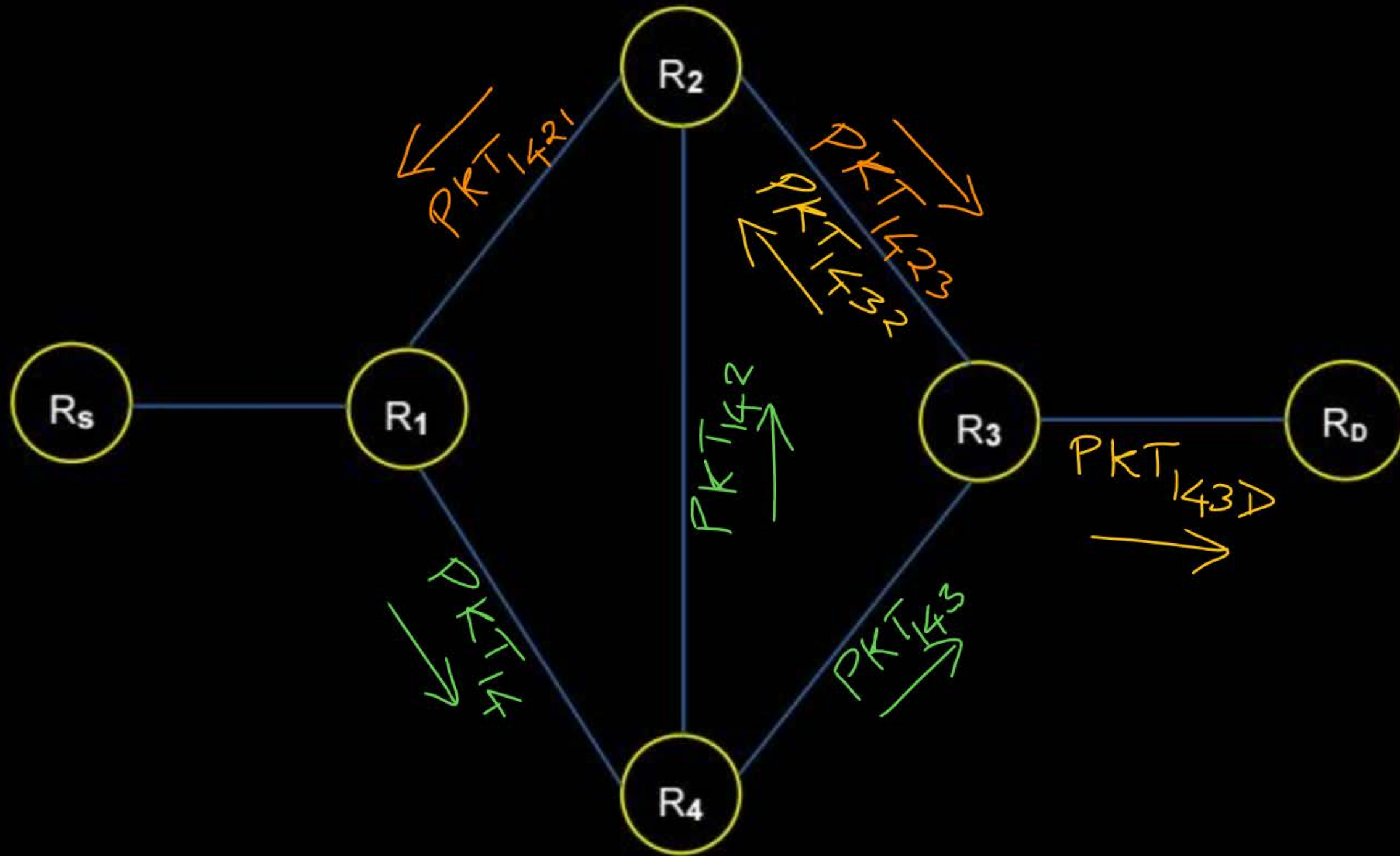


Topic : Flooding





Topic : Flooding



* ICMP Error : Time Exceeded

- ① Generated only when
dest. IP address is Host IP Address
(Unicast Transmission)
- ② Not generated when
dest. IP address is Broadcast IP Address
(Broadcast Transmission)



Topic : Dynamic Routing



→ Adaptive Routing

→ Types of Dynamic Routing :

1. Dijkstra's Link State Routing Algorithm

2. Bellman Ford Distance Vector Routing Algorithm

[MCQ]

H.W.

IIT-R

[GATE-2025] [1 Mark]



#Q. Consider the routing protocols given in List I and the names given in List II:

List I	List II
(i) Distance vector routing	(a) Bellman-Ford
(ii) Link state routing	(b) Dijkstra

For matching of items in List I with those in List II, which ONE of the following options is CORRECT?

- A (i) – (a) and (ii) – (b)
- B (i) – (a) and (ii) – (a)
- C (i) – (b) and (ii) – (a)
- D (i) – (b) and (ii) – (b)



Topic : Routing Protocols

- To exchange routing information between routers
- Two categories :
 1. Interior Gateway Protocol [IGP] ✓
 2. Exterior Gateway Protocol [EGP]



Topic : Interior Gateway Protocol

→ Intra-domain routing

→ within an autonomous system
(a network or group of networks)

[within ISP]

→ Routing Protocols :

1.1 Routing Information Protocol (RIP)

→ Distance Vector Routing Algo

[RIP-DV]

1.2 Open Shortest Path First (OSPF)

→ Link State Routing Algo.

[OSPF-LS] ✓



Topic : Exterior Gateway Protocol

- Inter-domain routing
- Between different autonomous system on the internet
- Routing Protocols :

[B/W ISPs]

2.1 Border Gateway Protocol (BGP) ✓

- Path Vector Routing Algo.

[MCQ] / H.W.

IIT-KGP

[GATE-2014] [1 Mark]



#Q. Which one of the following is TRUE about interior Gateway routing protocols - Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) ?

- ☐ A RIP uses distance vector routing and OSPF uses link state routing
- ☐ B OSPF uses distance vector routing and RIP uses link state routing
- ☐ C Both RIP and OSPF use link state routing
- ☐ D Both RIP and OSPF use distance vector routing



2 mins Summary



Topic

Routing

Topic

Static Routing



THANK - YOU

